

SOVABIDS: EEG-to-BIDS conversion software focused on automation, reproducibility and interoperability

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Summary

Electroencephalography (EEG) data are used in many fields, from neuroscience to clinical research, but recordings are stored in a wide variety of formats and lab-specific folder structures, making them difficult to organize, share, or compare across studies. SOVABIDS is an open-source tool that helps researchers convert EEG data into the Brain Imaging Data Structure (BIDS) ([Gorgolewski et al., 2016](#)), a standard that promotes FAIR data practices (Findability, Accessibility, Interoperability, and Reusability) and is increasingly being adopted for data sharing and large-scale collaboration. Rather than manually renaming files or writing dataset-specific scripts, SOVABIDS allows users to define conversion rules in human-readable YAML configuration files, which are then applied across the dataset ([Figure 1](#) illustrates this process). The tool is usable as a Python package, command-line tool, or an experimental interactive terminal user interface (TUI), and its API supports integration with graphical frontends for users who prefer a visual interface. Documentation and tutorials are available at sovabids.readthedocs.io.

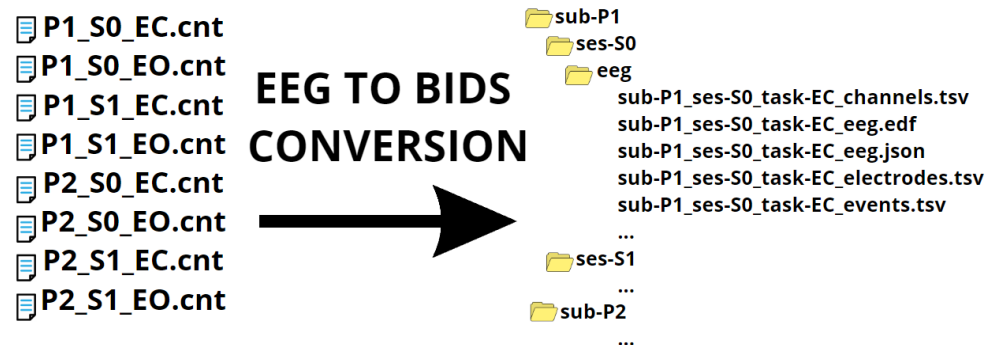


Figure 1: Illustration of the EEG to BIDS conversion. The left side shows raw EEG files with participant-specific naming conventions (for example, P1_S0_EC.cnt), where P1 and P2 represent participants, S0 and S1 indicate sessions, and EC (Eyes Closed) and EO (Eyes Open) refer to tasks. These raw files are converted into the BIDS format, shown on the right, where data are systematically organized into subject (sub-), session (ses-), and modality (eeg) folders. Each EEG recording is saved in standardized BIDS-compliant formats, including .edf for EEG signals and .tsv/.json for metadata.

29 Statement of need

30 Electroencephalography (EEG) is a widely used neuroimaging technique that provides
31 high temporal resolution for studying brain activity. Its applications span numerous
32 fields, including cognitive neuroscience, clinical diagnostics, brain-computer interfaces,
33 and neuroengineering. With the increasing volume and complexity of EEG data, ensuring
34 reproducibility, standardization, and interoperability has become a growing priority in the
35 field. The Brain Imaging Data Structure extension for EEG (EEG-BIDS) (Pernet et al.,
36 2019) provides a consistent framework for organizing EEG datasets, facilitating data sharing
37 (Markiewicz et al., 2021), collaboration, and cross-study comparison, and supporting FAIR
38 data practices (Wilkinson et al., 2016).

39 Despite these advantages, converting EEG datasets to BIDS in practice remains a significant
40 challenge. EEG data varies widely in format due to the multitude of proprietary standards used
41 by different hardware vendors, and datasets are often organized according to lab-specific or
42 acquisition-driven conventions rather than any consistent structure. For researchers without
43 strong programming backgrounds, particularly those at smaller or less well-resourced institutions,
44 this conversion process typically requires either substantial manual effort or the ability to write
45 dataset-specific scripts, both of which are error-prone and difficult to reproduce or scale to
46 large multi-participant studies. SOVABIDS addresses this gap by providing a rule-based,
47 semi-automated conversion workflow that is accessible to users with limited programming
48 experience while remaining flexible enough to handle the heterogeneity common in real-world
49 EEG datasets.

50 State of the Field

51 Several tools support EEG conversion to BIDS, including MNE-BIDS (Appelhoff et al., 2019),
52 data2bids in FieldTrip (Oostenveld et al., 2011), EEG-BIDS in EEGLAB (Delorme & Makeig,
53 2004), and EEG2BIDS (Rogers et al., 2022). More general-purpose converters such as Bidsme
54 (Beliy et al., 2023) also support EEG alongside other modalities.

55 MNE-BIDS (Appelhoff et al., 2019) provides a programmatic interface within the MNE
56 ecosystem and offers fine-grained control over metadata specification. However, it typically
57 requires dataset-specific scripting, which limits accessibility for users without programming
58 experience.

FieldTrip (Oostenveld et al., 2011) and EEGLAB (Delorme & Makeig, 2004) provide conversion utilities integrated within their respective analysis environments. While convenient for users already embedded in those ecosystems, conversion workflows often require manual interaction or scripting for each dataset.

EEG2BIDS (Rogers et al., 2022) offers a more guided workflow but relies on detailed user input at the file level, which becomes impractical for large heterogeneous datasets.

Bidsme (Beliy et al., 2023) is a general-purpose converter that assumes datasets are already organized in a structured hierarchy prior to conversion. While effective when data are consistently arranged, many EEG datasets are stored using vendor-specific naming schemes or lab-driven folder structures that do not conform to any standard hierarchy. In such cases, substantial manual reorganization is required before conversion can begin.

A natural question is whether SOVABIDS' goals could have been achieved by contributing to an existing tool, particularly MNE-BIDS. We argue they could not. MNE-BIDS is designed around a scripting paradigm where conversion logic is expressed in Python code. Adding rule-based, configuration-driven automation as a non-breaking extension would require architectural changes that diverge from MNE-BIDS' current scripting-oriented design philosophy and typical usage patterns. The same barrier applies to FieldTrip and EEGLAB, where conversion is tightly coupled to their respective analysis environments.

SOVABIDS instead introduces a distinct approach: an explicit two-tier separation between dataset-level rules and participant-level mappings, encoded in human-readable YAML configuration files. Like other tools, SOVABIDS requires dataset-specific configuration, but this configuration is defined once at the dataset level and then propagated to generate participant-level mappings, removing the need for per-file scripting. Combined with a flexible, semi-automatic API for extracting BIDS entities from file paths, this design allows SOVABIDS to handle the heterogeneous directory structures and vendor-specific naming conventions common in real-world EEG datasets. While the YAML-based approach introduces its own modest learning requirement, it offers a declarative alternative to general-purpose scripting.

Software Design

The central design tension in SOVABIDS is between expressiveness and accessibility: more expressive conversion logic typically requires programming skill, while simpler interfaces tend to sacrifice flexibility. The following five design principles reflect the trade-offs made to resolve this tension.

1. Accessibility for non-technical users

A scripting-based interface, as used by tools like MNE-BIDS, offers maximum expressiveness but requires users to write and maintain dataset-specific code. We traded this expressiveness for accessibility by using human-readable YAML configuration files instead. This approach was inspired by Bidscoin (Zwiers et al., 2022), a BIDS converter for MRI data. While YAML introduces its own learning requirement for users unfamiliar with the format, it offers a declarative configuration that is reusable across similar datasets and auditable without programming knowledge. The trade-off is that highly unusual conversion scenarios may require more verbose configuration, but for many EEG datasets the YAML-based approach is sufficient and more approachable than writing a custom script. To further lower the barrier to adoption, step-by-step guides and usage examples are provided in the documentation.

2. Automation that can accommodate outliers

EEG experiments typically produce multiple identically-organised datasets, one per participant. In practice, however, data organisation often varies slightly between participants due to

technical issues, partial recordings, or repeated segments. A fully automated system that assumes identical structure across participants would silently fail in these cases, while a fully manual system would not scale. SOVABIDS resolves this by separating conversion logic into two configuration files (illustrated in Figure 2):

- The **Rules File**, which encodes general conversion rules for the full dataset.
- The **Mappings File**, populated from the Rules File, which holds specific conversion parameters for every individual file.

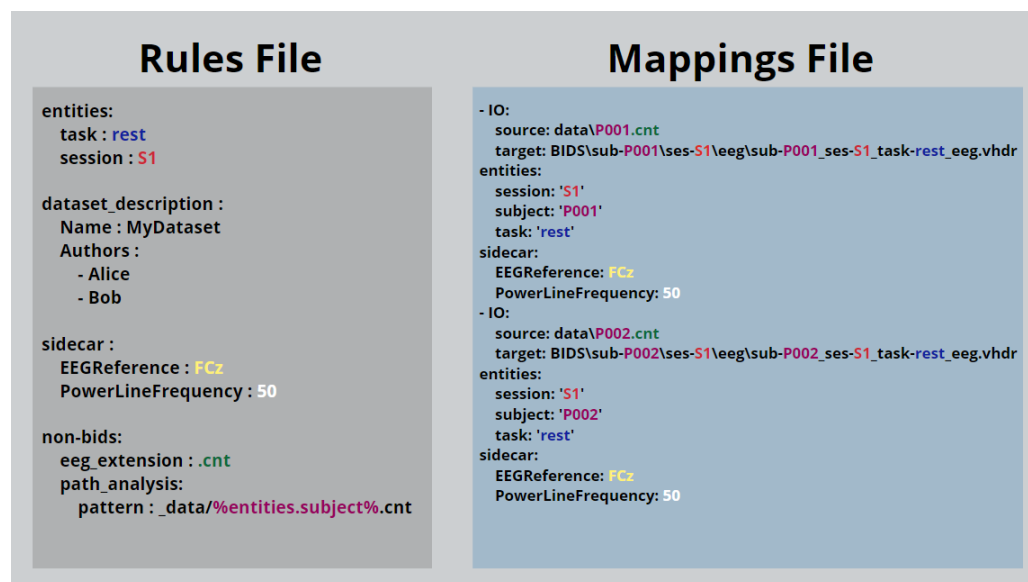


Figure 2: From a Rules File, a mapping for each file in the dataset can be generated and saved in the Mappings File. The colors illustrate how the information in both files is related.

This two-tier approach is inspired by tools such as Bidscoin (Zwiers et al., 2022) and HeuDiConv (Halchenko et al., 2023) (both focused on MRI). It generates a separate mapping for each file, which users can review and edit when a participant's data does not follow the general structure. Users can also start from an existing Rules File shared within a lab or community, and can connect an external GUI via SOVABIDS' API for supervised adjustment of edge cases where full automation is not possible.

To support flexible metadata extraction, SOVABIDS provides a semi-automatic API for inferring subject, session, task, and other BIDS-relevant properties from arbitrary file paths. Rather than requiring data to be pre-organized into a standard folder hierarchy before conversion can begin, as both Bidscoin and Bidsme do, SOVABIDS extracts these properties through pattern matching that is defined once at the dataset level and applied automatically across all files. This is supported through three approaches of increasing technicality: paired source-target examples, placeholder-based templates, and full regular expressions.

3. Reproducible conversion

Reproducibility requires that a conversion be fully specified by its saved configuration, so that repeating it does not depend on undocumented manual steps. The rules and per-file mappings are saved in the configuration files, and each run is logged. This allows users to audit, correct, and re-run conversions when a BIDS validator flags structural issues or when downstream analysis reveals incorrect metadata.

4. Accessible interfaces and interoperability

SOVABIDS provides two interactive access paths beyond the CLI and Python API, targeting different user needs and deployment environments.

The first is an experimental, optional terminal user interface (TUI), launched via the `sovatui` command (installed with the `sovabids[tui]` extra), which guides users through the full conversion workflow in four steps (Setup, Rules, Mappings, and Convert) without requiring any code. A TUI was chosen over web-based and native GUI alternatives for its portability, small dependency footprint, and lower maintenance burden. It runs wherever a terminal is available, including over SSH on HPC clusters and remote servers where large-scale EEG processing often takes place. It adds a single pure-Python dependency, the `textual` library, rather than the large binary dependencies of native toolkits such as Qt. Finally, it avoids the packaging pipelines, frontend toolchains, and platform-specific bugs that web and native GUIs accumulate. A walkthrough is available at <https://youtu.be/dOWiMTuGvAA>.

For external integration, SOVABIDS exposes an RPC-based API that lets external applications interact with its conversion logic. RPC was chosen over REST because its action-oriented design maps naturally onto the procedural steps of a data conversion workflow. To demonstrate the API's usability, a [reference web GUI was developed in Flask](#) and is available as a working [example](#). This path is intended for platforms that wish to embed SOVABIDS in a richer desktop or web frontend.

5. Format support through MNE delegation

Supporting the full range of EEG hardware formats from scratch would be an ongoing maintenance burden disproportionate to the tool's core contribution. SOVABIDS instead delegates file reading entirely to MNE-Python ([Gramfort et al., 2013](#)), and BIDS-compliant saving to MNE-BIDS ([Appelhoff et al., 2019](#)), inheriting support for all formats those libraries handle. The trade-off is that SOVABIDS' format coverage is bounded by MNE's, but this is an acceptable constraint given MNE's broad and actively maintained format support. In practice, SOVABIDS has been tested with BrainVision (.vhdr), EDF (.edf), EEGLAB (.set), and FIF (.fif) as input formats, converting each to BrainVision output by default (or FIF for MEG). Other formats that MNE can read (Neuroscan .cnt, BDF, KIT, and CTF) are handled through the same MNE delegation but are not independently tested in SOVABIDS' continuous integration suite. Basic MEG datatype routing is implemented, but MEG-specific BIDS requirements (empty-room recordings, manufacturer calibration files, and digitization coordinate systems) are not currently exposed through SOVABIDS' rule system and must be handled manually.

Architecture Overview

The five design principles above are reflected directly in SOVABIDS' two-module architecture, illustrated in [Figure 3](#). The Rules Module takes the user-defined Rules File and applies it across all EEG files in the dataset, extracting conversion parameters and compiling them into a Mappings File. This separation means that general conversion logic and participant-specific details are handled in distinct artifacts rather than embedded in code. The Conversion Module then reads the Mappings File and performs the actual transformation to BIDS-compliant output, delegating file reading to MNE and BIDS-compliant saving to MNE-BIDS. Users can interact with both modules through the CLI, the Python API, or the experimental TUI. At either stage, the RPC API additionally allows external tools and GUIs to inspect or modify the configuration, supporting the supervised adjustment workflows described above.

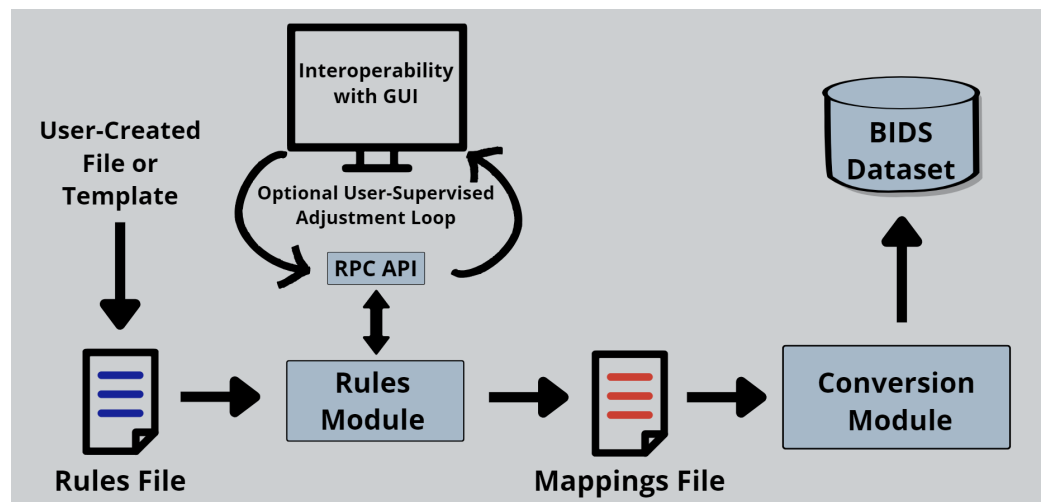


Figure 3: The architecture of SOVABIDS. The conversion process starts with a user-defined Rules File, which encodes general conversion rules (represented in blue inside the Rules File). The Rules Module processes these rules to generate a Mappings File, which contains specific configurations for all EEG files (each red line in the Mappings File represents the configuration of a different file). The Conversion Module then applies these configurations to produce a BIDS-compliant dataset. Interoperability is enabled via an RPC API, allowing integration with external tools, including graphical user interfaces for optional user-supervised adjustments.

Research Impact Statement

SOVABIDS is listed in the [official BIDS converter registry](#) under the EEG/MEEG/iEEG category. It is also available on the Neurodesk platform www.neurodesk.org (Dao et al., 2025; Renton et al., 2024), a community-maintained open neuroimaging environment, which broadens its accessibility beyond the development team.

Its use has been documented in peer-reviewed and academic work: a Master's thesis on EEG-based Alzheimer's risk classification (Henao Isaza, 2023), a Bachelor's thesis on web-based EEG processing tools (Zapata Saldarriaga, 2022), and a peer-reviewed study on harmonizing EEG features across multiple recording sites (Jaramillo-Jimenez et al., 2024).

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AI Usage Disclosure

The primary architecture and core functionality of this software were completed prior to December 2023. Generative AI tools were not used in the conceptual design, methodological decisions, or scientific development of the project.

198 After March 2025, limited use of generative AI tools was made for maintenance and supporting
199 tasks. The tools used were ChatGPT Codex (o4-mini) and ChatGPT (GPT-4.1) and Claude
200 (Sonnet 4.6).

201 AI assistance was used for updating and improving GitHub Actions continuous integration
202 workflows, enhancing and clarifying documentation, assisting in the implementation of a
203 utility function for generating random 1/f signals, minor code refactoring and formatting
204 improvements, and minor improvements on the paper.

205 All AI-assisted outputs were carefully reviewed, edited, tested, and validated by the authors.
206 All core design decisions and scientific judgments were made by the human authors.

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